ADMM CASE STUDY

HEALTHCARE PATIENT ANALYTICS & PREDICTION

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I. INTRODUCTION

1.1 Introduction:

Healthcare Patient Analytics & Prediction is a data analytics and research project developed to study patient admissions, demographic characteristics, medical conditions, clinical vital signs, treatment activity, and analytical risk classification.

The project uses healthcare datasets containing information about patients, doctors, hospital admissions, vital signs, and treatments. These datasets are processed and integrated to create a structured analytical environment for examining healthcare records.

The solution follows an ETL (Extract, Transform, Load) workflow using Python and Pandas in Google Colab. The processed datasets are exported as CSV files and loaded into a MySQL Data Warehouse. SQL is used for data validation and structured analysis, while Power BI Desktop is used to create interactive dashboards and visualize the research findings.

The project focuses on:

- Patient demographic analysis
- Hospital admission and discharge analysis
- Length-of-stay analysis
- Diagnosis-wise admission analysis
- Patient history and previous admission analysis
- Clinical vital-sign analysis
- Chronic-condition analysis
- Rule-based analytical risk classification
- Treatment and procedure analysis
- Doctor workload analysis
- Interactive healthcare analytics dashboards

1.2 Research Problem & Objectives:

- ### Research Problem

  Healthcare datasets contain information about patients, admissions, medical conditions, clinical measurements, and treatments. However, these records may be distributed across separate datasets and may require data cleaning, validation, integration, and transformation before they can be analyzed consistently.

Without an integrated analytical structure, it becomes difficult to examine:

- Patient and admission distributions
- Admission patterns over time
- Differences in length of stay
- Diagnosis-wise admission characteristics
- Relationships between previous admissions and analytical risk levels
- Variations in vital-sign measurements
- Chronic-condition distributions
- Treatment utilization across diagnoses and risk categories
- Relationships between demographic characteristics and risk classification
- Data quality and referential integrity across healthcare datasets

• Objectives-

- Load the available healthcare CSV datasets into Google Colab.
- Explore the datasets using Python and Pandas.
- Examine record counts, column structures, data types, missing values, duplicates, and unique identifiers.
- Validate primary keys and relationships between patients, doctors, admissions, treatments, and vital signs.
- Clean and transform the source datasets.
- Standardize date fields and derive patient age and length of stay.
- Analyze previous admissions and chronic-condition information.
- Aggregate vital-sign measurements at the admission level.
- Aggregate treatment activity at the admission level.
- Develop an integrated admission-level analytical dataset.
- Define a rule-based analytical risk score using selected patient and admission characteristics.
- Classify admissions into Low, Medium, and High analytical risk categories.
- Export the cleaned datasets as CSV files.
- Create a structured MySQL Data Warehouse.
- Load the transformed datasets into MySQL Workbench.
- Validate table record counts, primary keys, foreign keys, and data consistency.
- Perform SQL-based healthcare data analysis.
- Develop DAX measures for healthcare KPIs.
- Establish relationships between the imported Power BI tables.
- Develop interactive Power BI dashboards for admission, risk, clinical, and treatment analysis.
- Document the research findings, methodological limitations, and key project outcomes.

II. Architecture Overview

2.1 Architecture Overview

The Healthcare Patient Analytics & Prediction solution follows a structured ETL, Data Warehouse, SQL Analytics, and Business Intelligence architecture.

The workflow begins with the source healthcare datasets and progresses through data exploration, validation, cleaning, transformation, analytical feature engineering, CSV staging, MySQL loading, SQL analysis, Power BI data modeling, DAX development, and dashboard creation.

1. Source Data

The project uses 5 datasets:

- `patients.csv`
- `doctors.csv`
- `admissions.csv`
- `vitals.csv`
- `treatments.csv`

These datasets contain patient demographics, doctor information, admission records, clinical measurements, and treatment details.

2. Python / Google Colab

Python and Pandas were used for:

- Loading the source datasets
- Exploring dataset structures
- Examining data types and record counts
- Identifying missing values and duplicates
- Validating unique identifiers and foreign keys
- Converting date fields
- Calculating patient age and length of stay
- Processing chronic-condition information
- Aggregating clinical vital signs
- Aggregating treatment records
- Creating admission-level analytical features
- Developing the rule-based risk score and risk categories
- Exporting the cleaned datasets as CSV files

3. ETL & Cleaned Data

The ETL stage transforms the source records into structured datasets suitable for analytical storage.

Major processing activities included:

- Replacing missing chronic-condition values with No Recorded Condition
- Converting date fields to datetime format
- Calculating length of stay
- Calculating patient age using the latest admission date as the reference date
- Calculating previous admissions for each patient
- Creating a chronic-condition indicator
- Separating systolic and diastolic blood pressure values
- Aggregating vital-sign measurements by admission
- Aggregating treatment activity by admission
- Combining patient, admission, vital-sign, and treatment features
- Creating analytical risk scores and risk levels
- Validating the final analytical dataset

4. MySQL Data Warehouse

The transformed datasets were loaded into the MySQL database:

healthcare_analytics_dw

The warehouse contains six tables:

- dim_patient
- dim_doctor
- fact_admission
- fact_treatment
- fact_vital
- fact_admission_analytics

The patient and doctor tables provide descriptive information. The admission, treatment, and vital tables retain detailed healthcare records, while the admission analytics table stores aggregated clinical features and analytical risk classifications.

5. SQL Research Analysis

SQL was used for:

- Overall admission analysis

- Patient count and admission distribution
- Length-of-stay analysis
- Monthly admission analysis
- Diagnosis-wise analysis
- Risk-level distribution
- Age-group and chronic-condition analysis
- Previous-admission analysis
- Vital-sign analysis
- Treatment utilization analysis
- Doctor workload analysis
- Primary-key and foreign-key validation
- Final analytical dataset consistency checks

6. Power BI

Power BI Desktop was used for:

- Connecting to the MySQL Data Warehouse
- Importing the six analytical tables
- Establishing table relationships
- Creating DAX measures
- Developing KPI cards
- Building line, bar, column, donut, and stacked-column charts
- Comparing admission and risk distributions
- Examining clinical vital-sign patterns
- Comparing treatment utilization
- Creating four interactive dashboard pages

The four dashboard pages are:

1. Executive Overview
2. Patient Risk & Readmission Analysis
3. Clinical & Vital Signs Analysis
4. Treatment & Hospital Operations Analysis

2.2 Architecture Flow



III. PROJECT APPROACH

3.1 Methodology

Five source workbooks were identified:

- patients.csv
- doctors.csv
- admissions.csv
- treatments.csv
- vitals.csv

The source files were loaded into Google Colab using Python and Pandas. Their structures, column names, data types, identifiers, date fields, and missing values were examined before preprocessing.

Initial Dataset Size

The source dimension tables included customer, employee, geography, product, reseller, and sales territory data. The two sales datasets contained Internet and Reseller transactions.

Dataset	Records	Columns
Patients	1,690	8
Doctors	40	5
Admissions	3,324	7
Treatments	8,329	6
Vitals	33,148	7

Data Extraction & Exploration

The datasets were explored in Google Colab using Python and Pandas. The analysis included:

- Record counts and column names
- Data types and numerical statistics
- Missing-value analysis
- Duplicate-record and primary-key validation
- Date-range analysis
- Patient age and length-of-stay calculations
- Diagnosis and treatment distributions
- Vital-sign summaries

- Patient, doctor, admission, treatment, and vital-record relationship checks

Data Validation

Validation checks were performed for:

- Duplicate records and duplicate primary keys
- Missing values and inconsistent data types
- Patient and doctor references in admission records
- Admission references in treatment and vital records
- Admission and discharge date consistency
- Treatment and vital timestamps in relation to admission periods
- Final analytical dataset completeness

No full-row duplicates or duplicate primary-key identifiers were found. Foreign-key validation identified zero invalid references across the checked patient, doctor, admission, treatment, and vital relationships.

The admission, treatment, and vital datasets contained timestamps outside their associated admission periods. These records were retained, and the limitation was documented rather than removing records solely because of their timestamps.

Data Transformation

The transformation process included:

- Converting date and time columns to appropriate data types
- Handling missing chronic-condition information using the value No Recorded Condition
- Calculating patient age and admission length of stay
- Deriving previous-admission counts
- Splitting blood-pressure readings into systolic and diastolic values
- Aggregating vital-sign measurements at admission level
- Aggregating treatment, procedure, and medication counts
- Combining patient, doctor, admission, vital, and treatment information
- Creating admission-month, age-group, and risk-related fields
- Applying a rule-based risk classification using age, previous admissions, chronic-condition status, and vital-sign flags

The risk classification was created for analytical comparison. It does not represent a clinically validated diagnosis or a trained readmission-prediction model.The final admission-level analytical dataset, healthcare_analytics.csv, contained **3,324 records and 32 columns**, with no missing values and no duplicate admission identifiers.

Data Staging & Loading

The cleaned and transformed datasets were exported from Google Colab as CSV files. The final exported files included:

- patients_clean.csv
- doctors_clean.csv
- admissions_clean.csv
- treatments_clean.csv
- vitals_clean.csv
- healthcare_analytics.csv

These files were imported into MySQL Workbench after the database schema and table structures were created.

MySQL Data Warehouse

The cleaned analytical tables were loaded into:

healthcare_analytics_dw

The database contains the following tables:

- dim_patient
- dim_doctor
- fact_admission
- fact_treatment
- fact_vital
- fact_admission_analytics

Power BI Data Modeling & Reporting

The MySQL data warehouse was connected to Power BI Desktop. Relationships were configured between patient and admission data, doctor and admission data, admission and treatment data, admission and vital data, and admission and admission-level analytical data.

DAX measures, KPI cards, charts, and other visuals were developed across four report pages:

- 01_Executive_Overview
- 02_Risk_Analysis
- 03_Clinical_Vitals
- 04_Treatment_Operations

3.2 Technology Stack

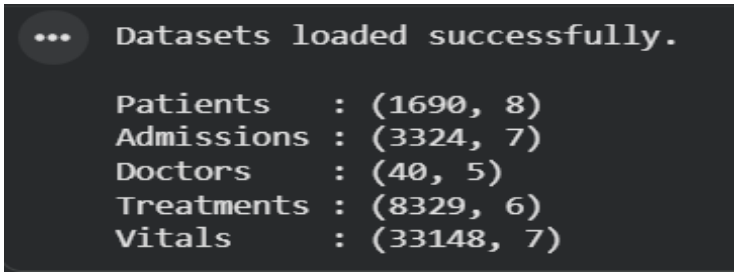
Component	Technology	Purpose
Source data	CSV files	Patient, doctor, admission, vital-sign and treatment records
Data processing	Python, Pandas, NumPy	Data exploration, cleaning and transformation
Development environment	Google Colab	ETL, feature engineering and analytical preparation
Data staging	CSV	Transfer of transformed datasets
Data warehouse	MySQL	Structured analytical storage
Database tool	MySQL Workbench	Database creation, data loading and SQL queries
Data analysis	SQL	Healthcare data validation and research analysis
Visualization	Power BI Desktop	Interactive healthcare dashboards
Calculations	DAX	KPI measures and analytical comparisons
Documentation	Microsoft Word	Research documentation

IV. SOURCE DATA & TABLE MAPPING

4.1 Source Data

The project uses five primary source datasets.

Source file	Purpose
patients.csv	Patient demographics, date of birth, gender, address and chronic-condition information
doctors.csv	Doctor identifiers, names and specializations
admissions.csv	Admission and discharge dates, diagnoses, patient references and doctor references
vitals.csv	Recorded heart rate, blood pressure, oxygen level and temperature
treatments.csv	Treatment dates, procedures, medications and dosages



4.2 MySQL Table Mapping

MySQL table	Source or transformed dataset	Purpose
dim_patient	patients_clean.csv	Patient demographic and chronic-condition details
dim_doctor	doctors_clean.csv	Doctor and specialization details
fact_admission	admissions_clean.csv	Detailed admission and discharge records
fact_treatment	treatments_clean.csv	Treatment and medication records
fact_vital	vitals_clean.csv	Detailed clinical measurements
fact_admission_analytics	healthcare_analytics.csv	Aggregated admission features and analytical risk classification

V. DATA EXPLORATION & PRIMARY KEY VALIDATION

5.1 Data Exploration

Initial data profiling was performed using Python and Pandas.

The analysis included:

- Record counts and column names
- Data-type inspection
- Missing-value analysis
- Full-row duplicate analysis
- Primary-key uniqueness checks
- Numerical descriptive statistics
- Date-range analysis
- Admission and discharge validation
- Blood pressure parsing
- Diagnosis-wise admission counts
- Chronic-condition distribution
- Treatment and medication distribution
- Vital-sign aggregation
- Foreign-key validation

The patient dataset contained 240 missing chronic-condition values. No full duplicate rows or duplicate identifiers were identified in the five source datasets.

5.2 Data Quality and Relationship Validation

MySQL table	Records	Unique primary-key identifiers
dim_patient	1,690	1,690
dim_doctor	40	40
fact_admission	3,324	3,324
fact_admission_analytics	3,324	3,324
fact_treatment	8,329	8,329
fact_vital	33,148	33,148

5.3 Admission and Patient Analysis

The dataset contained 3,324 admission records associated with 1,690 distinct patients.

The overall admission-level length of stay had the following characteristics:

Metric	Result
Total admissions	3,324
Unique patients	1,690
Average length of stay	5.97 days
Minimum length of stay	2 days
Maximum length of stay	10 days
Admissions with stay of 7 days or more	1,448
Percentage with stay of 7 days or more	43.56%

The recorded admission dates ranged from 11 March 2025 to 11 March 2026.

5.4 Diagnosis Distribution

The admission records contained seven diagnosis categories.

Diagnosis	Admissions	Percentage
Hypertension	746	22.44%
Respiratory Infection	689	20.73%
Chest Pain	485	14.59%
Heart Failure	473	14.23%
Pneumonia	454	13.66%
Asthma Attack	252	7.58%
Diabetes Complication	225	6.77%
Total	**3,324**	**100% approximately**

5.5 Vital-Sign Analysis

Vital signs were aggregated at the admission level to calculate average and selected minimum or maximum values.

Vital-sign measure	Average
Heart rate	89.86 bpm

Oxygen level	93.98%
Temperature	99.01 °F
Systolic blood pressure	130.11 mmHg
Diastolic blood pressure	82.45 mmHg

A project-defined vital-risk flag was created when an admission's average measurements met at least one of these conditions:

- Average heart rate greater than 100
- Average oxygen level below 92
- Average temperature greater than 100 °F
- Average systolic blood pressure greater than 140
- Average diastolic blood pressure greater than 90

The resulting distribution was:

Vital-risk flag	Admissions	Percentage
No flag	2,977	89.56%
At least one flag	347	10.44%
Total	**3,324**	**100%**

These thresholds were used for project analysis and are not a substitute for clinical assessment.

5.6 Rule-Based Risk Classification

The rule-based risk score was applied to all 3,324 admission records.

Risk level	Score range	Admissions	Percentage
Low	0–2	1,400	42.12%
Medium	3–4	1,443	43.41%
High	5–7	481	14.47%
Total	**0–7**	**3,324**	**100%**

The mean length of stay by analytical risk category was:

Risk level	Average length of stay
Low	5.94 days
Medium	5.96 days
High	6.05 days

The classification represents the score-based categories defined for this project. Since no observed readmission outcome was available, the risk categories cannot be used to calculate a true readmission rate or establish predictive accuracy.

VI. MYSQL DATA WAREHOUSE IMPLEMENTATION & DATA LOADING

6.1 MySQL DataBase Implementation

After data cleaning and transformation, the datasets were loaded into the MySQL database:

healthcare_analytics_dw

MySQL Workbench was used for database creation, table loading, SQL analysis, record-count validation, and referential integrity checks.



Figure 6.1: Tables created in MySQL Workbench

6.2 Data Warehouse Structure

- **Dim_doctor**

doctor_id	first_name	last_name	specialization	contact_no
301	Anthony	Reese	Pulmonologist	5965377606
302	Erica	Owens	Pulmonologist	9212316949
303	Brittany	Cisneros	Cardiologist	4695711788
304	Tracy	James	Neurologist	9451082387
305	Mary	Vazquez	Neurologist	8459209544

- **Dim_patient**

doctor_id	first_name	last_name	specialization	contact_no
301	Anthony	Reese	Pulmonologist	5965377606
302	Erica	Owens	Pulmonologist	9212316949
303	Brittany	Cisneros	Cardiologist	4695711788
304	Tracy	James	Neurologist	9451082387
305	Mary	Vazquez	Neurologist	8459209544

- **Fact_vital**

vital_id	admission_id	recorded_time	heart_rate	blood_pressure	oxygen_level	temperature	systolic_bp	diastolic_bp
5001	2001	2026-03-06 20:06:42	120	129/81	94	99.94	129	81
5002	2001	2026-03-02 18:52:52	118	130/71	96	97.13	130	71
5003	2001	2026-02-10 11:50:10	65	134/73	91	100.22	134	73
5004	2001	2026-01-05 10:58:43	77	121/70	94	100.48	121	70
5005	2001	2026-02-18 22:47:00	86	131/85	95	99.80	131	85

- **Fact_treatment**

treatment_id	admission_id	treatment_date	procedure	medication	dosage
7001	2001	2025-05-12 00:00:00	IV Fluid Therapy	Atorvastatin	10mg daily
7002	2001	2025-12-20 00:00:00	IV Fluid Therapy	Atorvastatin	50mg daily
7003	2001	2025-07-28 00:00:00	IV Fluid Therapy	Atorvastatin	25mg daily
7004	2001	2025-07-06 00:00:00	IV Fluid Therapy	Atorvastatin	50mg daily
7005	2002	2026-02-07 00:00:00	ECG Monitoring	Aspirin	25mg daily

- **Fact_admission_analytics**

admission_id	patient_id	admission_date	discharge_date	diagnosis	doctor_id	room_no	length_of_stay	previous_admissions	age	gender	chronic_conditions	has_chronic_condition	avg_heart_rate	min_heart_rate	max_heart_rate
2001	1001	2026-02-07 00:00:00	2026-02-13 00:00:00	Diabetes Complication	317	R418	6	0	49	Female	Diabetes	1	91.00	65	120
2002	1001	2026-03-07 00:00:00	2026-03-17 00:00:00	Heart Failure	320	R396	10	1	49	Female	Diabetes	1	90.92	67	117
2003	1002	2025-05-16 00:00:00	2025-05-21 00:00:00	Asthma Attack	339	R411	5	0	61	Female	Asthma	1	83.87	61	118
2004	1003	2025-12-08 00:00:00	2025-12-10 00:00:00	Respiratory Infection	325	R120	2	1	86	Female	No Recorded Condition	0	83.78	60	114
2005	1003	2025-08-11 00:00:00	2025-08-21 00:00:00	Pneumonia	306	R368	10	0	86	Female	No Recorded Condition	0	90.40	66	114

- **Fact_admission**

admission_id	patient_id	admission_date	discharge_date	diagnosis	doctor_id	room_no	length_of_stay	previous_admissions
2001	1001	2026-02-07 00:00:00	2026-02-13 00:00:00	Diabetes Complication	317	R418	6	0
2002	1001	2026-03-07 00:00:00	2026-03-17 00:00:00	Heart Failure	320	R396	10	1
2003	1002	2025-05-16 00:00:00	2025-05-21 00:00:00	Asthma Attack	339	R411	5	0
2004	1003	2025-12-08 00:00:00	2025-12-10 00:00:00	Respiratory Infection	325	R120	2	1
2005	1003	2025-08-11 00:00:00	2025-08-21 00:00:00	Pneumonia	306	R368	10	0

avg_oxygen_level	min_oxygen_level	avg_temperature	max_temperature	avg_systolic_bp	avg_diastolic_bp	vital_readings	treatment_count
93.89	88	99.25	100.48	129.44	83.00	9	4
92.67	88	99.04	100.85	119.92	84.58	12	1
94.20	88	98.62	100.41	133.33	82.13	15	4
92.22	88	98.91	100.60	139.56	79.56	9	2
95.00	89	99.20	100.86	134.10	81.50	10	2

procedure_count	medication_count	admission_month	vital_risk_flags	age_group	risk_factor_count	risk_score	risk_level
1	1	2026-02	0	46-60	1	2	Low
1	1	2026-03	0	46-60	2	3	Medium
1	1	2025-05	0	61-75	1	3	Medium
1	1	2025-12	0	76+	1	4	Medium
1	1	2025-08	0	76+	0	3	Medium

VII. POWERBI DATA MODEL & KPI DEVELOPMENT

7.1 Power BI Data Model

Power BI Desktop was connected to the healthcare_analytics_dw MySQL Data Warehouse.

All six tables were imported into the Power BI model. The model connects patient and doctor dimensions to admission records, while treatment and vital-sign records are connected through admission identifiers. The admission-level analytical table is linked to the admission table using the admission identifier.

The model supports analysis across patient demographics, doctors, admission characteristics, detailed treatments, clinical measurements, and analytical risk classifications.



Figure 7.1: Power BI Data Model

7.2 POWER BI RELATIONSHIPS

The relationships were configured with appropriate cardinality and filter direction to support accurate reporting.

The following relationships were configured:

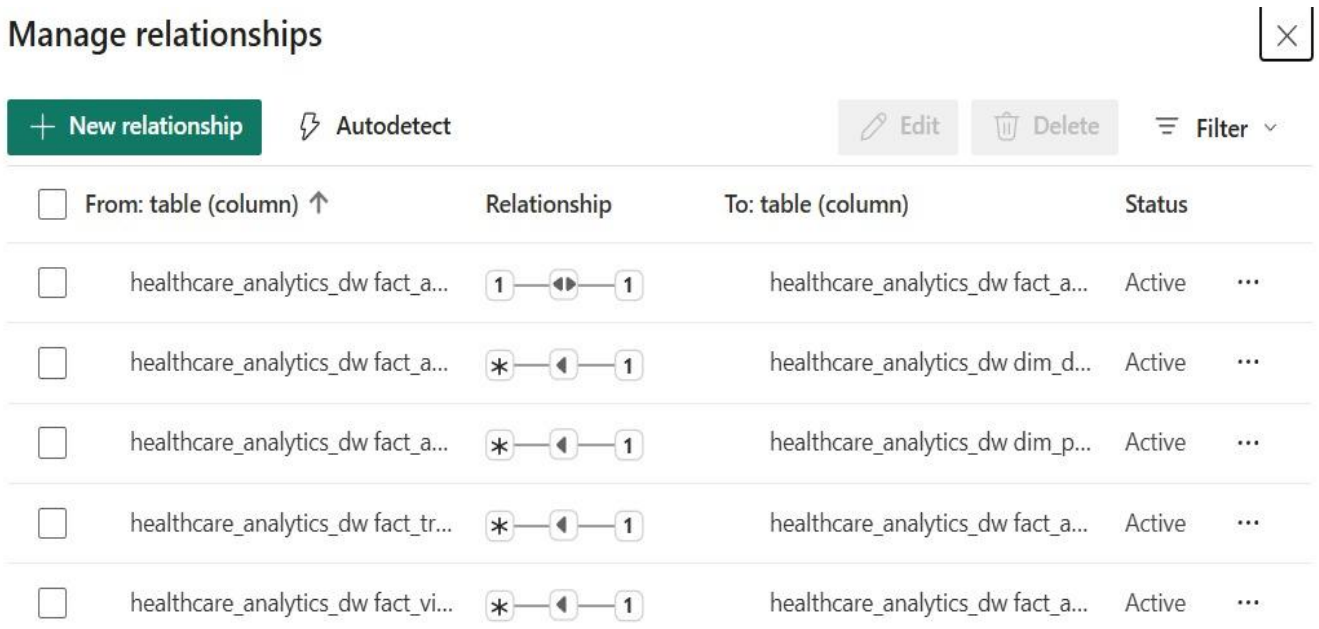
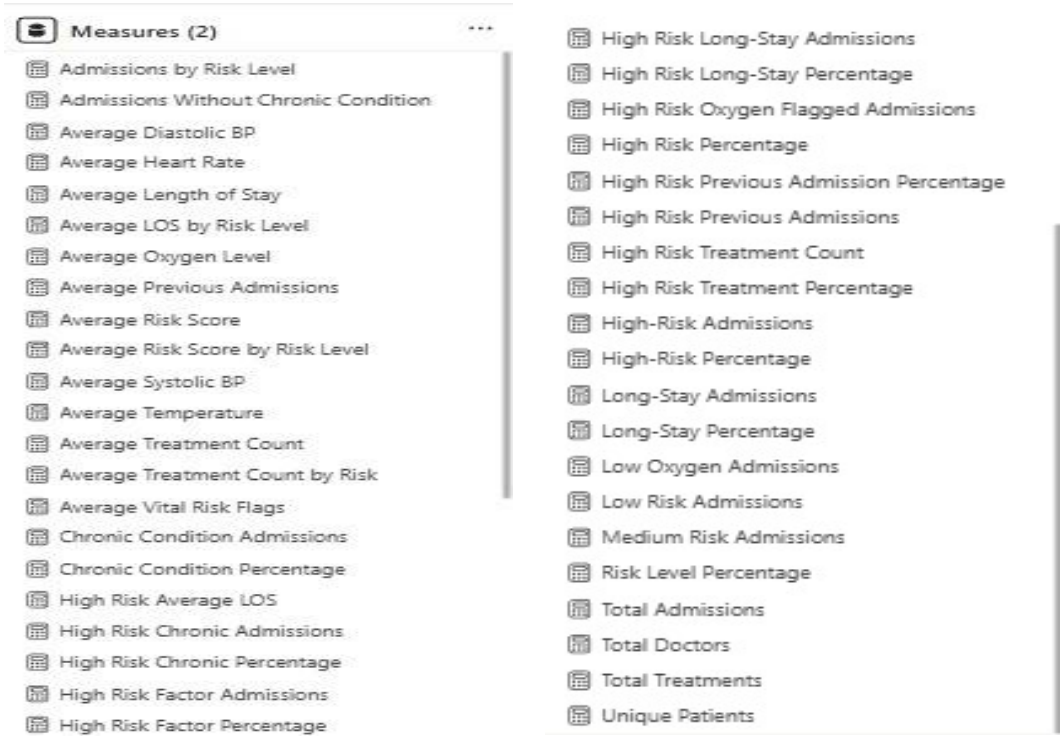


Figure 7.2: Power BI Table Relationships

7.3 POWER BI MEASURES



VIII. Power BI Dashboards & Key Insights

8.1 Executive Overview Dashboard

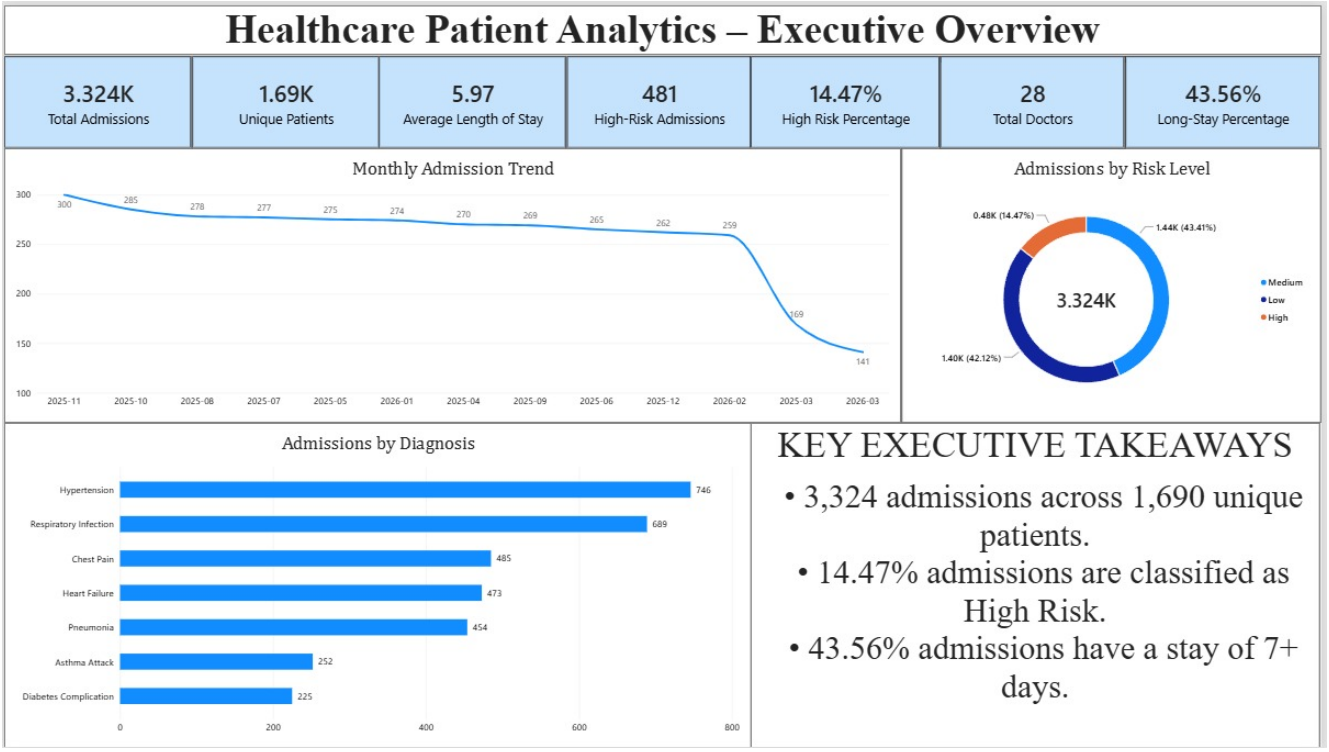


Figure 8.1: Executive Overview Dashboard

Key Insights:

- This page presents the overall admission and patient summary.
- It includes KPI cards for total admissions, unique patients, average length of stay, high-risk admissions, high-risk percentage, long-stay percentage, and active doctors.
- The visuals include the monthly admission trend, admission distribution by risk level, and admissions by diagnosis.

8.2 Patient Risk & Readmission Analysis

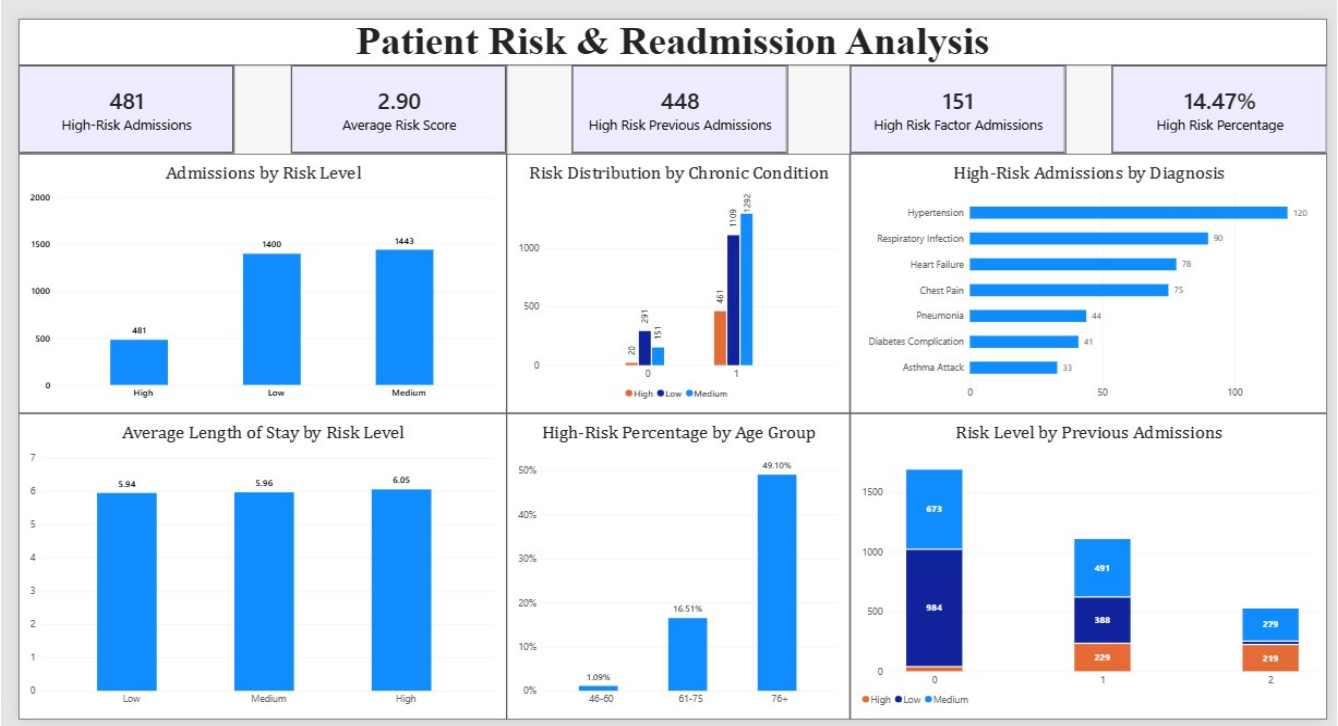


Figure 8.2: Patient Risk & Readmission Analysis

Key Insights:

This page examines the project-defined analytical risk classifications.

It includes risk-level KPI cards and visuals for:

- Admissions by risk level
- Average length of stay by risk level
- High-risk admissions by diagnosis
- Risk level by previous admissions
- Risk distribution by chronic condition
- High-risk percentage by age group

The page presents the distribution of rule-based risk categories and their relationship with selected patient and admission characteristics.

8.3 Clinical & Vital Signs Analysis

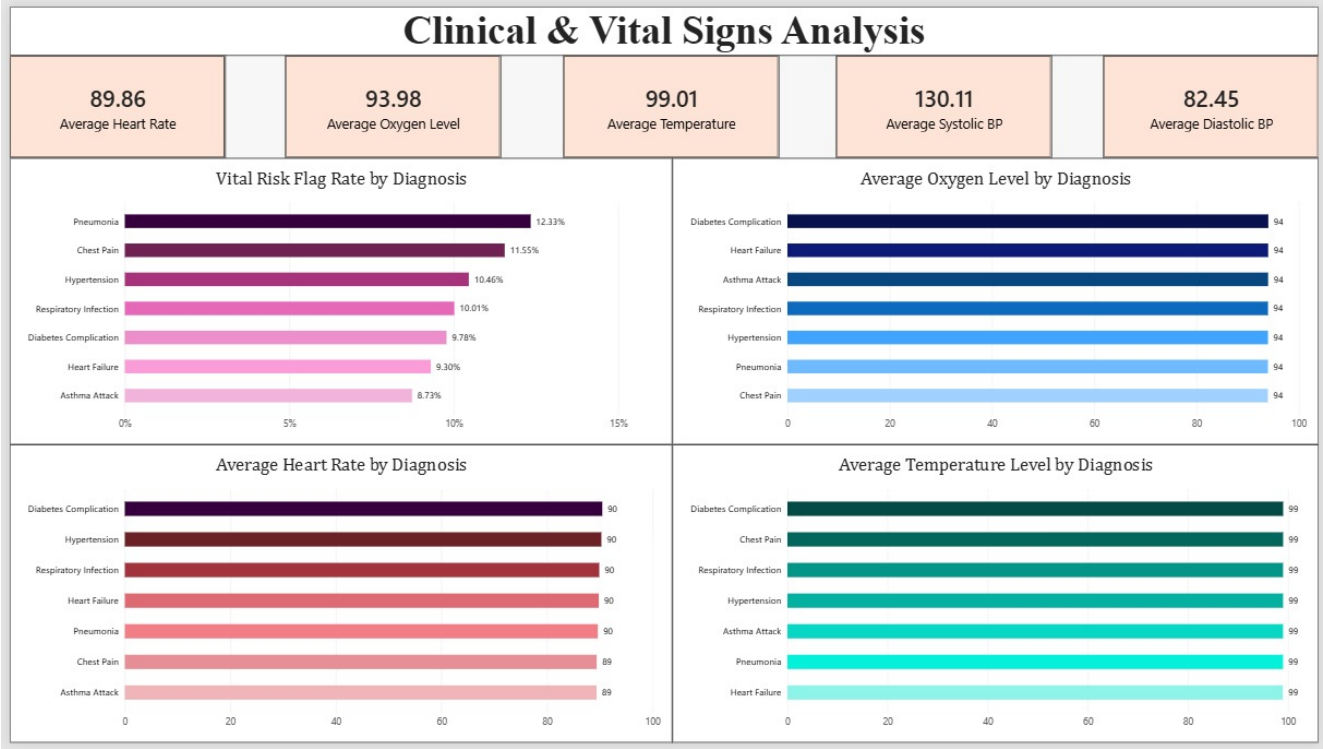


Figure 8.3:Clinical & Vital Signs Analysis Dashboard

Key Insights:

- This page focuses on aggregated vital-sign measurements and project-defined vital-risk flags.
- It includes KPI cards for average heart rate, oxygen level, temperature, systolic blood pressure, and diastolic blood pressure.
- The visuals compare vital-risk flag rates, average heart rate, average oxygen level, and average temperature across diagnosis categories.

8.4 Treatment & Hospital Operations Analysis Dashboard

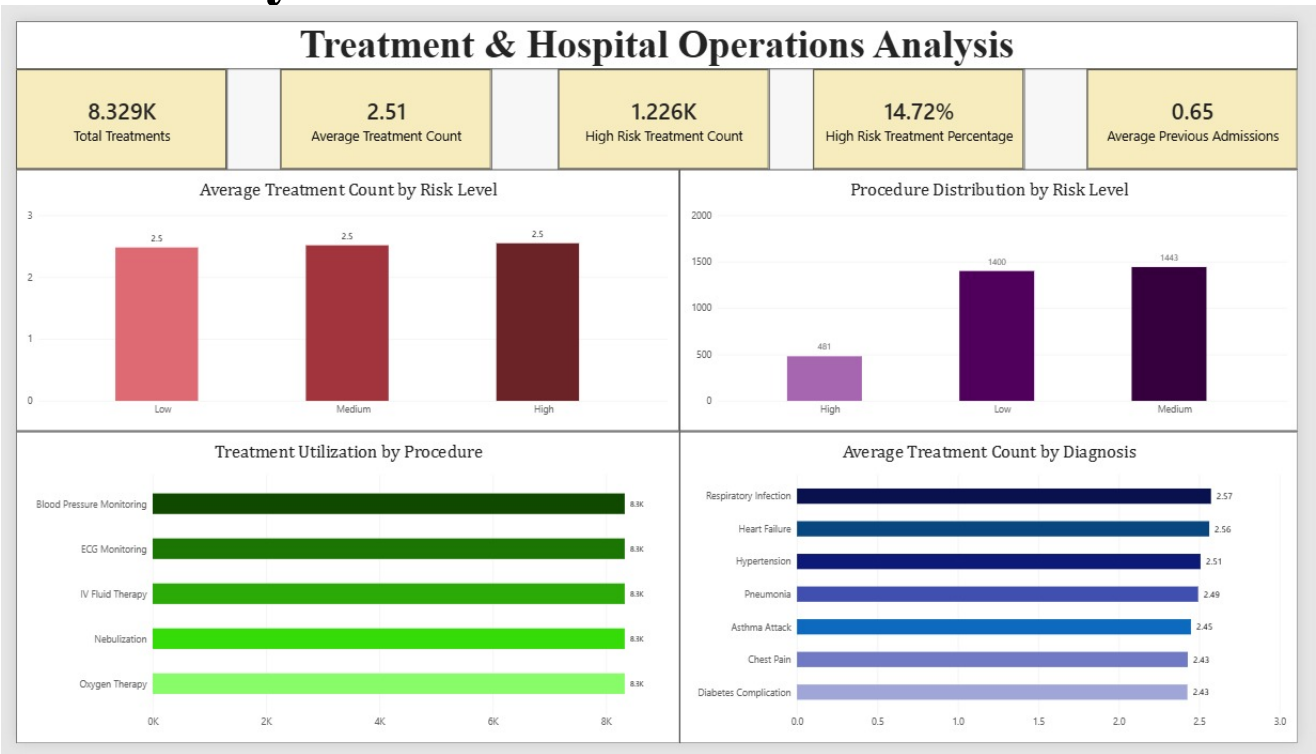


Figure 8.4:Treatment & Hospital Operations Analysis Dashboard

Key Insights:s

- This page examines treatment counts, procedure utilization, and treatment patterns across analytical risk categories and diagnoses.
- It includes KPI cards for total treatments, average treatment count, high-risk treatment count, high-risk treatment percentage, and average previous admissions.
- The visuals include average treatment count by risk level, treatment utilization by procedure, procedure distribution by risk level, and average treatment count by diagnosis.

IX. CONCLUSION

9.1 Conclusion

The Healthcare Patient Analytics & Prediction case study developed an end-to-end healthcare analytics workflow using Python, Google Colab, MySQL Workbench, SQL, Power BI, and DAX.

Five healthcare datasets were cleaned, validated, and transformed into a structured MySQL Data Warehouse containing six tables. An admission-level analytical dataset with 3,324 records and 32 columns was created, incorporating patient details, vital-sign aggregates, treatment counts, and rule-based risk classifications.

SQL analysis and four Power BI dashboards were used to examine admission trends, diagnoses, length of stay, clinical measurements, risk categories, and treatment utilization.

The project demonstrates ETL, data validation, relational modeling, feature engineering, SQL analysis, and interactive visualization. The risk classification remains exploratory, as validated readmission outcomes were unavailable for predictive model training and evaluation.

9.2 Key Outcomes

- Implemented a healthcare analytics workflow using Google Colab, Python, MySQL Workbench, SQL, and Power BI Desktop.
- Loaded and examined five healthcare source datasets.
- Identified and handled 240 missing chronic-condition values.
- Validated source identifiers and the tested relationships between datasets.
- Created cleaned and transformed datasets for all five source entities.
- Built a MySQL Data Warehouse containing six analytical tables.
- Integrated admission, patient, vital-sign, and treatment features into an admission-level dataset containing 3,324 records and 32 columns.
- Derived patient age, length of stay, previous admissions, chronic-condition indicators, and aggregated clinical measurements.
- Developed a rule-based analytical risk score ranging from 0 to 7.
- Classified 1,400 admissions as Low risk, 1,443 as Medium risk, and 481 as High risk under the defined scoring rules.
- Developed DAX measures for admission, length-of-stay, risk, clinical, and treatment analysis.
- Built four Power BI dashboard pages covering executive overview, analytical risk classification, clinical vital signs, and treatment operations.

- Documented data-quality limitations related to treatment and vital-sign timestamps.
- Identified the absence of an observed readmission outcome as a limitation for predictive modeling.

9.3 GitHub Link

https://github.com/surabhi-0405/Case-Study-8-Healthcare-Patient-Analytics-Prediction